5.6 Composition of Functions

Name _____ Period ____ Date _____

Algebra II 5.6 (30-minute plan) · Student sheet

I can…

- evaluate a composition of functions.
- find the composition of two functions and its domain.
- use composition to solve real-life problems.

Vocabulary · fill in the missing word

Composition — f(g(x)): the _____ of g is the input of f

Domain of f(g(x)) — the _____ in the domain of g whose outputs g(x) are in the domain of f

Part 1 · Notes and examples (fill in each note, then work the example)

Composition: f(g(x)) means: find _____ first, then use that output as the _____ of f.

Composition: Work from the _____ out.

Example 1

Let $f(x) = 5x - 1$ and $g(x) = x^2 + 2$. Find $f(g(-1))$ and $g(f(-1))$.

Order matters: f(g(x)) and g(f(x)) are usually _____.

Example 2

Let $f(x) = 3x - 4$ and $g(x) = x^2$. Find $f(g(x))$ and $g(f(x))$.

Domain of f(g(x)): The domain of f(g(x)): x must be in the domain of _____, and g(x) must be in the domain of _____.

Example 3

Let $f(x) = \sqrt{x}$ and $g(x) = x - 5$. Find $f(g(x))$ and its domain.

Part 2 · Partner practice

Partner practice

Let $f(x) = x^2 - 1$ and $g(x) = 2x + 3$. Find $f(g(x))$.